

Neuroimaging / Optimal neuroimaging measures for early detection

MRI-based automated volumetric segmentation tool in the detection of early Alzheimer's disease

Wanting Liu¹ | Lisa W.C. Au² | Jill Abrigo³ | Yishan Luo⁴ | Adrain Wong¹ |
 Pauline Kwan² | Alison Hon Wing Ma¹ | Anthea Yee Tung Ng⁵ | Sirong Chen⁶ |
 Eric Y.L. Leung⁶ | Chi Lai Ho⁶ | Winnie C.W. Chu³ | Ho Ko⁵ | Lin Shi³ |
 Vincent C.T. Mok²

¹ Gerald Choa Neuroscience Centre, Therese Pei Fong Chow Research Centre for Prevention of Dementia, Lui Che Woo Institute of Innovative Medicine, Chinese University of Hong Kong Prince of Wales Hospital, Shatin, Hong Kong

² Lui Che Woo Institute of Innovative Medicine, Hong Kong, China

³ The Chinese University of Hong Kong, Shatin, Hong Kong

⁴ BrainNow Research Institute, Shenzhen, China

⁵ Gerald Choa Neuroscience Centre, Therese Pei Fong Chow Research Centre for Prevention of Dementia, Lui Che Woo Institute of Innovative Medicine, The Chinese University of Hong Kong Prince of Wales Hospital, Hong Kong, China

⁶ Hong Kong Sanatorium & Hospital, Happy Valley, Hong Kong

Correspondence

Wanting Liu, Gerald Choa Neuroscience Centre, Therese Pei Fong Chow Research Centre for Prevention of Dementia, Lui Che Woo Institute of Innovative Medicine, Chinese University of Hong Kong, Prince of Wales Hospital, Shatin, Hong Kong.
 Email: yuentinglau@gmail.com

Abstract

Background: Cognitive unimpaired (CU) and mild cognitive impairment (MCI) subjects harboring beta-amyloid (A+) and tau (T+) are at high-risk for incident cognitive decline. Fluoro-deoxyglucose (FDG) positron emission tomography (PET) likely represents both cumulative loss of the neuropil and functional impairment of neurons while magnetic resonance imaging (MRI) indicates cumulative loss and shrinkage of the neuropil. As PET and CSF analysis in detecting beta-amyloid and tau are not easily accessible (either costly or invasive nature or radiation), we compared the performance of MRI-based automated segmentation tool with FDG PET in identifying CU and MCI subjects harboring A+T+.

Method: A total of 62 subjects (CU=37, MCI=25) underwent MRI, FDG PET, ¹¹C- PIB and ¹⁸F-T807 PET. MRIs were processed by an automated segmentation tool to obtain an Alzheimer's Disease (AD) - resemblance atrophy index (AD-RAI) which was derived from a machine learning method (AccuBrain®) that analyzed multiple brain regions relevant to AD.

Result: AD-RAI yielded the same sensitivity (0.73) among all subjects with FDG PET while they had similar specificity. AD-RAI performed the highest sensitivity (0.91) in MCI subjects over FDG PET (0.82) with an acceptable specificity (0.79) which was slightly lower than that of FDG PET. However, both AD-RAI and FDG PET were sub-optimal in CU group. Detailed results are demonstrated in the Table.

Conclusion: The volumetric segmentation tool achieves a good diagnostic profile similar to FDG PET in identifying early AD. It has the potential to be more widely used as a tool to diagnose early AD for receiving treatment or recruitment into clinical trials.

TABLE 1
Table. Sensitivity, specificity and accuracy of AD-RAI and FDG PET
(All subjects, n=62; MCI, n=25; CU, n=37)

		Sensitivity (95% CI)	Specificity (95% CI)	Accuracy
AD-RAI	All	0.73 (0.45-0.91)	0.92 (0.79-0.97)	88.71%
	MCI	0.91 (0.57-1.00)	0.79 (0.49-0.94)	84.00%
	CU	0.25 (0.01-0.78)	1.00 (0.87-1.00)	91.89%
FDG PET	All	0.73 (0.45-0.91)	1.00 (0.91-1.00)	93.55%
	MCI	0.82 (0.48-0.97)	1.00 (0.73-1.00)	92.00%
	CU	0.50 (0.09-0.91)	1.00 (0.87-1.00)	94.59%

AD-RAI=Alzheimer's disease resemblance atrophy index; FDG PET=Fluoro-deoxyglucose positron emission tomography; MCI=mild cognitive impairment; CU=cognitively unimpaired.